

Indus Institute of Technology & Engineering
Indus University
CSE/CE/IT
Subject: Cryptography and Network Security(CE0634)
End Sem Exam Question Bank(Unit 1-4)

UNIT- 1

PYQ

Ratto mari kadhoo OM's IMP

1. Explain all key objectives of cryptography.
2. Draw and Explain OSI Architecture.
3. Differentiate symmetric and asymmetric encryption?
4. Define Active and Passive attacks with Example.
5. Write a short note on Security Services.
6. Explain **playfair cipher** substitution technique in detail. Find out cipher text for the following given key and plaintext.
Key = ENGINEERING
Plaintext=COMPUTER SCIENCE AND ENGINEERING
7. Encrypt using HILL CIPHER the P.T “ATTACK CHINAA” using the key
GYDNSKURT
8. Differentiate between **Monoalphabetic and Polyalphabetic cipher**.
9. Explain the **Vigener Cipher** with an example.
10. Explain the Vernam Cipher with an example.
11. **Explain Block cipher mode of operation. and explain any 3 of them !**
12. Why is it important to study Feistel cipher?
13. Explain a single round function in DES.
14. Explain Byte substitution and Shift row operation of AES in detail.
15. Elaborate AES encryption with neat sketches.
16. Explain avalanche effect in DES and discuss strength of DES in brief.
17. Explain function of S-BOX in DES and also Explain avalanche effect.
18. What are the different Types of Attack on Encrypted Messages.
19. Write a short note on **Blowfish** with a diagram.
20. Specify the difference **between diffusion and confusion.**
21. Differentiate between **a block cipher and a stream cipher.**
22. Define **meet-in-the-middle attack.**
23. Explain Transposition Technique.
24. Write a short note on Triple DES.
25. Discuss the following terms in brief:
i) brute force attack ii) cryptography

Q.caeser cipher + Example and all

UNIT- 2

1. Define Asymmetric Key Encryption with a proper Diagram.
2. Explain RSA public key cryptography with all steps.
3. Explain Diffie Hellman Key Exchange with all steps.
4. Justify Diffie Hellman Key Exchange vulnerable to Man in Middle Attack. 5. P and Q are two prime numbers. $P=7$, and $Q=17$. Take public key $E=5$. If the plaintext value is 6, then what will be the cipher text value according to the RSA algorithm? Explain in detail.
6. Solve this example Using Knapsack algorithm
Super Increasing sequence= $\{1, 2, 4, 10, 20, 40\}$ Do necessary calculation to solve.
7. Explain in detail about Elliptic Curve Architecture.
8. Write a short note on Man in the Middle Attack.
9. Write a short note on Key Distribution and also define Master and Session Key.
10. List four general categories of schemes for the distribution of public keys. 11. Write a note on simple secret key distribution. Q10+12 = 10 marks
12. Draw a neat sketch showing the key distribution scenario.
13. Specify the application of public key cryptography.

Unit 3

1. Explain MACs based on Hash functions.
2. Discuss SHA-1 with proper diagrams.
3. Explain HMAC with a diagram.
4. Explain Elgamal and Schnorr Digital Signature algorithm with examples.
5. What is Kerberos and its requirements?
6. What is PGP? How authentication and confidentiality is maintained in PGP.
7. Explain authentication services of X.509.
8. Explain MD5 algorithm with diagram.
9. Explain NIST Digital Signature algorithm.
10. Differentiate between SHA-1 and MD5.

Unit 4

1. What is IP and Web Security? Explain the importance and features of IPSec.
2. How does IPSec work? Explain the process of IPSec.
3. List down various applications and benefits of IPSec.
4. Explain IPSec Architecture in detail.

5. Describe different types of IPSec Protocols.
6. Explain different types of Web Security Threats in detail.
7. What is the goal of system level security? Explain different types of system threats.
8. What is the working of Intrusion Detection System? Explain the types of intrusion detection systems.
9. What is Virus? List down and describe different types of virus.
10. What is a threat? Explain common Information Security Threats and its solutions.
11. Explain Virus Countermeasures.
12. What are the Characteristics of Firewalls? Describe the need and importance of firewalls.
13. Explain Firewall Design Principles in detail.
14. Explain the importance of Trusted Systems with examples.
15. What is SSL? Explain different protocols of SSL and types of SSL Certificates.
16. What is SET? Explain the functionality of SET in detail.

Dear Student, this Question bank is just for your reference. Do not expect the same question in Exam. Example values may be different in exam so prepare accordingly.